

(a) Construct three paired training samples

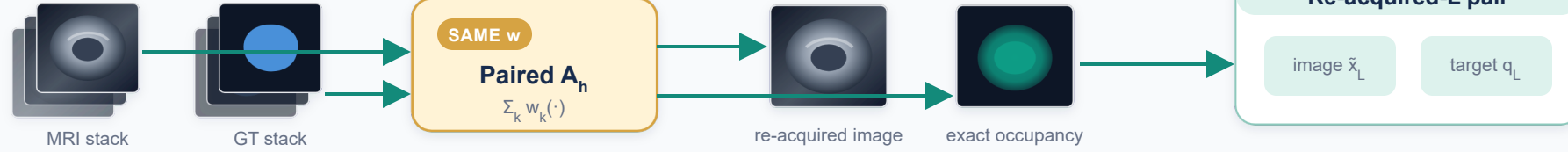
NATIVE LABELED

Identity path preserves native anatomy



RE-ACQUIRED L

The same profile transforms image and exact mask

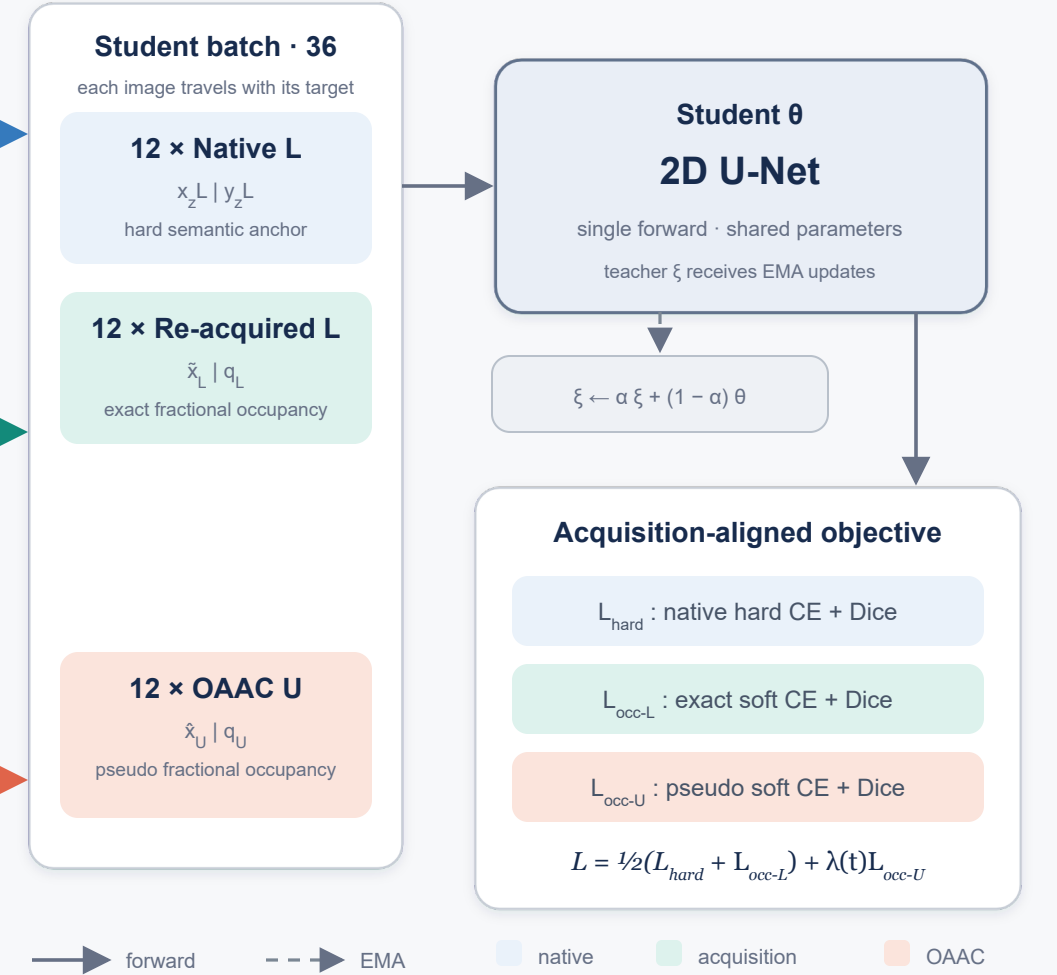


UNLABELED U

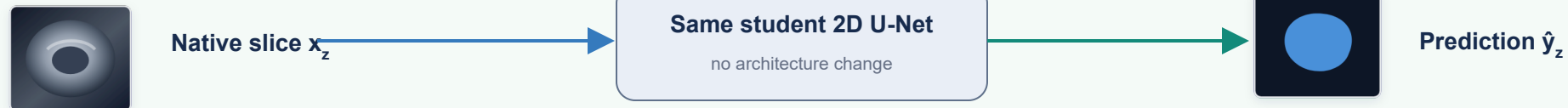
Acquisition first; target-preserving appearance second



(b) Shared optimization



(c) Inference $\cdot$ unchanged single-slice path



Training-only operations are removed

no neighboring slices $\cdot$ no A_h $\cdot$ no OAAC
no extra parameters or inference latency